

5)

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	5	-1	1	2
2	4	-1	3	$\frac{3}{2}$
3	3	-1	4	1
4	2	-1	5	1
5	1	0	6	$-\frac{1}{2}$
6	2	1	4	-2

Part 1) Given $h_1(x) = f(x) + g(x)$, find $h_1'(2)$ Part 2) Given $h_2(x) = f(x) - g(x)$, find $h_2'(3)$ Part 3) Given $h_3(x) = f(x) \cdot g(x)$, find $h_3'(4)$ Part 4) Given $h_4(x) = \frac{f(x)}{g(x)}$, find $h_4'(2)$ Part 5) Given $h_5(x) = (f(x))^2$, find $h_5'(2)$ Part 6) Given $h_6(x) = f(g(x))$, find $h_6'(6)$